Biochemistry Cume 12-1-2017

Proteins

1. How would you describe the terms "primary", "secondary", "tertiary", and "quaternary" structure of proteins? (20%)

2. Identify and distinguish the PrP^c (cellular) and PrP^{sc} (Scrapie) forms of prion protein. Explain why they have different biological functions even though they share the same amino acid sequence. Based on your answer, comment on the implications for computational prediction of protein structure (15%)

3. Myoglobin contains 8 alpha helices, one of which has the following sequence:

   QGAMN KALEH FRKDI AA

   Which side chains are likely to be on the side of the helix that faces the interior of the protein? Which are likely to be facing the aqueous solvent? Account for the spacing of the residues facing the interior. (15%)

4. Fetal hemoglobin (Hb F) contains serine in place of cationic histidine 143 of the beta chain of adult hemoglobin (Hb A). Residue 143 faces a central cavity between the beta chains.

   a) Why does 2,3-bisphosphoglycerate (BPG) bind more tightly to deoxy Hb A than to deoxy Hb F? (10%)
   b) How does the decrease of the affinity of Hb F for 2,3-BPG affect the affinity of Hb F for O_2? (10%)
   c) The P_{50} of Hb F is 18 torr, and the P_{50} of Hb A is 26 torr. How do these values explain the efficient transfer of oxygen from maternal blood to fetal blood? (10%)

5. Circular dichroism measurements have shown that poly-L-lysine is a random coil at pH 7, but becomes alpha helical as the pH is raised above 10. Account for this pH dependent conformational transition. Also predict the pH dependence of a helix-coil transition of poly-L-glutamate. (20%)